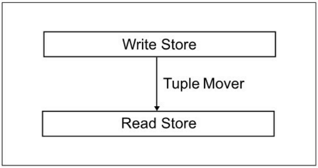
transactions (deposits and withdrawals) and credit-card transactions. OLAP workloads, on the other hand, typically perform complicated and compute-intensive analysis on the data stored in the database. For example, the CEO of a supermarket chain wants to know what the total volume of sales over the Christmas holiday period is, from each store. This requires OLAP query processing. OLAP queries are typically read-intensive. An OLTP transaction typically accesses one or a few records, but it accesses all fields of the accessed records. On the other hand, OLAP queries access a large number of records of the database, but only access one, or a very few, of the fields of each record.

The differences in the very nature of OLTP and OLAP workloads demand certain design differences in the databases for each of these workloads. An example of these differences is the row-oriented record layout of Row-Store Databases, versus the columnar layout of Column-Store Databases. In row stores, all the fields of the record are stored contiguously, with one record completely laid out before another record starts. In columnar stores, the column fields are laid out contiguously. Recall that OLTP typically operates on one or a very few number of records, and hence benefits from the row layout. On the other hand, OLAP workloads access a large number of records, accessing only one or a small number of columns in each record. Hence they benefit from the columnar layout. With a column-store architecture, a DBMS needs only to read the values of the columns required for processing a given query, and can avoid bringing into memory irrelevant fields that are not needed for the current query.

In warehouse environments where typical queries involve aggregates performed over large numbers of data items, a column store has a sizeable performance advantage over a row store. A number of popular column oriented databases have been developed, such as VoltDB (*www.voltdb.com)* and *Vertica (www.vertica.com).*

As mentioned earlier, OLAP is read-intensive; the database performance needs to be optimised for read operations, but at the same time, it needs to support database updates while maintaining ACID consistency. Some modern databases try to optimise for both, by providing both read- and write-optimised stores. C-store is the academic project on which the popular database analytics product, Vertica, is based. C-store provides both a read-optimised column store and an update/insert-oriented writeable store, connected by a tuple mover. C-Store supports a small Writeable Store (WS) component, which provides extremely fast insert and update operations. It also supports a large Read-optimised Store (RS), which is well optimised for query operations. Read-Store supports only a very restricted form of insert, namely the batch movement of records from WS to RS, a task that is performed by the tuple mover.

While till now the buzz in the database industry has been about 'Big Data', which denotes the huge volumes of data expected to be generated by peta-scale computing, it is not just the data that matters. After all, DBMS and data analytics are not just about raw data; they are about converting data into information, which is then converted by analytics tools into business insights. Therefore, the term 'Big Data' inherently covers the following requirements as well:



C-Store

1. Any future database system should be able to handle huge volumes of data (as denoted by 'Big Data' needs).
2. They should be able to process and query the data at great speeds (hence the need for fast data). This requirement brings forth the need for complex programming models that can query and process large data, such as Map-Reduce.
3. They should be able to analyse the data and convert it into business intelligence using complex analytics (hence the need for deep analytics).

Therefore the challenge for future database systems is not just about big data, but about handling *big data that requires fast query responses and deep analytics.*

Taking advantage of emerging hardware

It is not only the database industry that is undergoing many changes to address the twin challenges of big data and cloud computing. The hardware industry is at an inflection point today as we discuss below.

In 1965, Dr Gordon Moore from Intel predicted that the transistor density of semiconductor chips would double approximately every 18 to 24 months – this is popularly referred to as Moore's Law. For many decades, till now, such an exponential growth in the number of transistors on the chip has roughly translated into improvements in processor performance. Moore's Law enabled processor performance, more specifically single-thread performance, to double every 18 months—thereby allowing software developers to deliver increasingly complex functionality and greater performance without having to rewrite their code.

Though Moore's Law itself continues to hold good even now, the diminishing returns from hardware ILP extraction techniques, and the constraints on increasing the clock frequencies due to the limitations of power consumption and heat dissipation of the chip have forced chip manufacturers to look towards multi-core processor designs to improve